EE3032 Introduction to VLSI Design—Homework 3

1. Design a 4-bit serial-parallel multiplexer-based adder and verify the adder using Hspice. Measure the worst-case delay and power of the adder by applying the following stimuli (X, Y)=(1, 5); (15, 13), (11, 9), (4, 8), (3, 7), (12, 14), (2, 15), (6, 10). Also, report the total area of the adder in terms of $\sum_{i=0}^{n-1} W_i Li$, where W_i and L_i denote the width and length of the ith transistor in the adder. Design objective is to minimize the product of delay and power. (15 points) (Metrics of grade: energy and area)

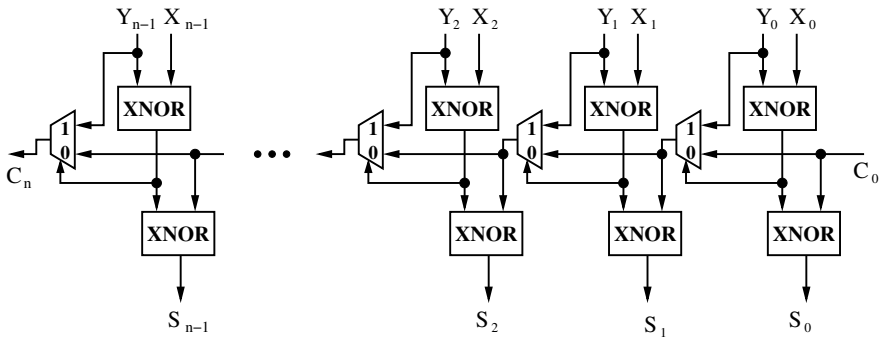

Figure 1: Block diagram of an n-bit multiplexer-based adder.